

COBURG UNIVERSITY OF APPLIED SCIENCES



M.Engg in Autonomous Driving

Test Strategy Document

Submitted by:

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1. Feature Name

Smooth and Seamless Autonomous Driving

2. Feature Owner

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3. Feature Description

Smooth and Seamless Autonomous Navigation is a system-level feature designed to enable the autonomous shuttle to travel continuously and reliably between requested locations within the Model City environment. (Smooth refers to the generation of smooth path for Motion execution and Seamless refers to the execution of trajectory control without interruptions)

From the user's perspective, the shuttle navigates between locations in a predictable and uninterrupted manner while safely executing straight paths and turning maneuvers. The navigation behavior remains continuous without unnecessary stops, ensuring a reliable and comfortable travel experience.

Internally, this feature combines path planning and trajectory control. Path planning generates a feasible and continuous navigation path between the requested start and destination locations, while lateral and longitudinal control ensures that the shuttle follows this path safely through appropriate steering and speed commands.

When a turning maneuver cannot be completed using forward-only motion due to vehicle kinematic constraints, the system performs a controlled reverse maneuver to safely complete the navigation task. The feature ensures predictable and collision-free navigation within the defined operational scope of the Model City environment.

A. Key Performance Indicators (KPI's)

The following KPIs define the system-level quality measures used to evaluate this feature:

- Planning success rate: A valid and continuous navigation path is generated for the requested destination.

KPI 1: Planning Success Rate

Total navigation requests executed: n

Successful path generations: n

Planning success rate: (expected 100%)

- Successful route completion: The shuttle reaches the destination without collision or unintended interruption.

KPI 2: Successful Route Completion

Total route executions: n

Successful destination arrivals: n

Route completion success rate: (expected 100%)

- Maneuver completion success: Turning maneuvers are completed safely within vehicle constraints.

KPI 3: Maneuver Completion Success

Reverse maneuver success: n

Steering constraint violations ($>30^\circ$): Expected 0

Maneuver completion success rate: (Expected 100%)

4. Feature Functionality / Properties

This feature provides the following functionality:

- Generates feasible navigation paths between requested locations.
- Produces continuous paths consisting of straight and turning segments.
- Executes the planned path using lateral and longitudinal control.
- Tracks navigation paths continuously using steering and speed commands.
- Executes forward-only motion when turning constraints allow.
- Executes controlled reverse maneuvers when forward-only motion is infeasible.
- Ensures continuous navigation behavior within the Model City layout.

5. Architecture

Input

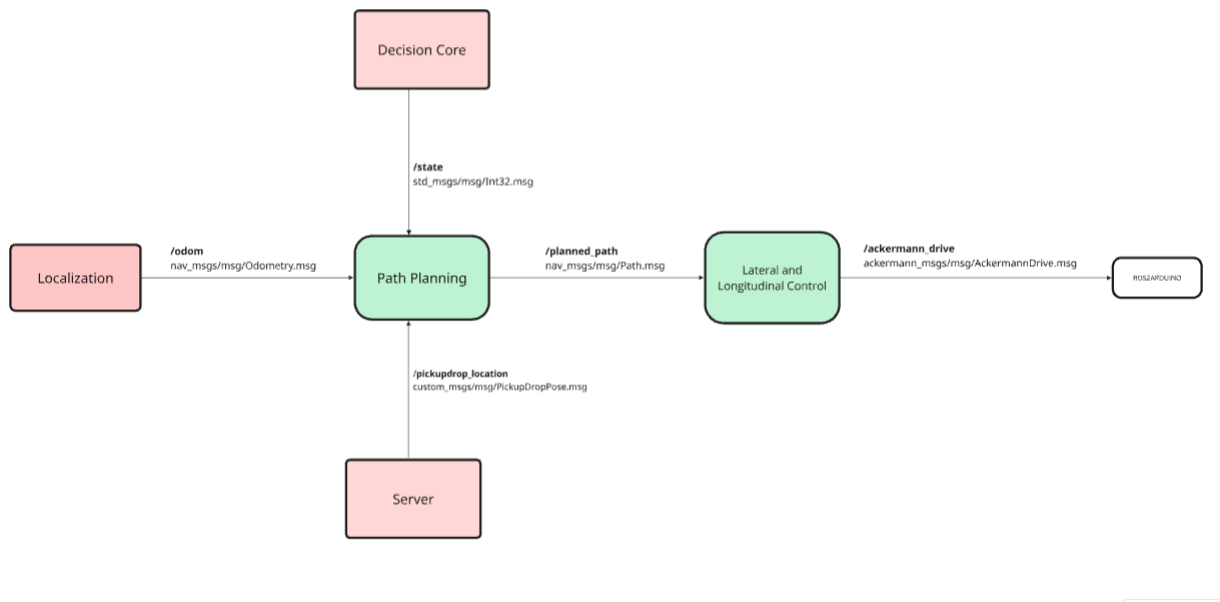
- **Navigation request (pickupdrop_location)**
The server publishes the requested pickup and drop-off locations. This input is used by the path planning to compute the navigation path between the requested locations
- **Vehicle pose and orientation (/odom)**
The localization module publishes vehicle position and orientation as odometry data. This information is required by the path planning module to determine the current vehicle position and generate a feasible path from the current pose to the destination.

Intermediate System Output

- **Planned navigation path (/planned_path)**
The path planning module publishes a continuous navigation path containing waypoints from the current vehicle position to the destination. This path is used as the reference trajectory for motion execution.

Output

- **Steering and speed commands (/ackermann_drive)**
The lateral and longitudinal control module generates steering angle and speed commands based on the planned path and current vehicle pose. These commands are sent to the Ros2arduino for motion execution.



6. Test Strategy

A. Test Basis

The system test cases for this feature are derived from the following test basis artifacts:

- Refined system-level requirements related to autonomous navigation behavior.
- Path planning and trajectory execution design specifications.
- System architecture documentation describing interaction between navigation planning and control execution.
- Expected vehicle-level behavior during autonomous navigation in the Model City environment.
- Feature test scenarios defined in the feature testing specification.

These artifacts are used to derive expected results and to ensure traceability between requirements and system test cases.

B. Test Analysis

Test analysis focuses on identifying operational situations in which navigation behavior may be affected by path geometry or system availability.

The primary risk is associated with turning maneuvers where vehicle kinematic limits prevent completion using forward-only motion. These situations require validation that the system safely performs a controlled reverse maneuver and continues navigation without collision.

Additionally, situations where localization becomes unavailable are analyzed to verify that

the system behaves safely and deterministically when required inputs for navigation are missing.

The analysis identifies the navigation conditions requiring validation and forms the basis for defining test scenarios and coverage criteria.

C. Test Techniques for Defining Test Cases

System testing is performed using scenario-based testing to verify system behavior under realistic operational conditions.

Scenario-based testing is selected because the feature represents an end-to-end navigation behavior experienced by the user. The testing approach validates both path generation and trajectory execution during complete navigation tasks.

The test design includes:

- A happy flow scenario representing nominal navigation behavior.
- An alternate flow scenario representing constrained navigation conditions requiring reverse maneuvers.
- An edge case scenario representing unsupported operational conditions due to missing localization input.

7. Scenarios for Testing

The following scenarios define the high-level system situations under which the KPIs are evaluated.

Scenario 1: Straight Path and 180° Turn with 4 m Radius (Happy Flow) KPI 1&2

The shuttle performs autonomous navigation between requested locations where the navigation path consists of a straight segment or a 180° turn with a feasible turning radius of 4 m. The system generates a continuous path and executes the maneuver using forward-only motion. The shuttle completes navigation without collision or interruption and reaches the destination successfully.

Scenario 2: 90° Turn and 180° Turn with 3 m Radius (Alternate Flow) KPI 1,2&3

The shuttle performs autonomous navigation requiring turning maneuvers that include a 90° turn and a constrained 180° turn with a radius of 3 m. Due to vehicle kinematic limitations, forward-only motion is insufficient to complete the maneuver. The system identifies the constraint and executes a controlled reverse maneuver to safely complete the turn. Navigation continues after the maneuver, and the destination is reached successfully without collision.

Scenario 3: Localization Loss During Motion Execution (Edge Case) NO KPI

The shuttle performs autonomous navigation between requested locations where localization is available at the start, and the navigation path is generated and executed normally. During motion execution, localization input becomes unavailable or is lost. Since valid vehicle pose information is required for trajectory execution, the system stops the vehicle and halts further motion. The system remains in a safe and deterministic state without generating motion commands, therefore it is an edge case and not a failure case since the feature cannot be tested as pre-condition is not met.

1. Test Coverage

Test coverage is defined to ensure that all relevant aspects of autonomous navigation behavior are validated at system level.

Requirements coverage:

All system-level requirements related to path generation, trajectory execution, maneuver handling, and safe system behavior during missing localization are covered by at least one system test case. The target requirement coverage is 100%.

Scenario coverage:

The defined scenarios cover nominal navigation behavior, constrained turning behavior requiring reverse maneuvers, and edge-case behavior involving missing localization input. This ensures validation of both supported and unsupported operational conditions.

KPI coverage:

The defined KPIs (planning success rate, successful route completion, maneuver completion success) are evaluated primarily within the alternate flow scenario, as this scenario represents the highest-risk operational condition involving constrained turning behavior and controlled reverse maneuvers during navigation.

In the happy flow scenario with continuous localization availability, KPI observations serve as a baseline reference for nominal system behavior and expected navigation performance.

In the edge case scenario, KPI evaluation is not applicable, as the scenario is intended to validate safe and deterministic system behavior under unsupported conditions rather than navigation performance.

E. Test Environment

System testing for the feature Smooth and Seamless Autonomous Navigation is conducted in the Model City test environment.

The environment consists of a structured road layout containing straight segments, 90° turns, and 180° turns with defined radii of 4 m and 3 m. All scenarios are executed using the ROS 2-based autonomous shuttle system and visualization tools to observe navigation behavior and system outputs.

